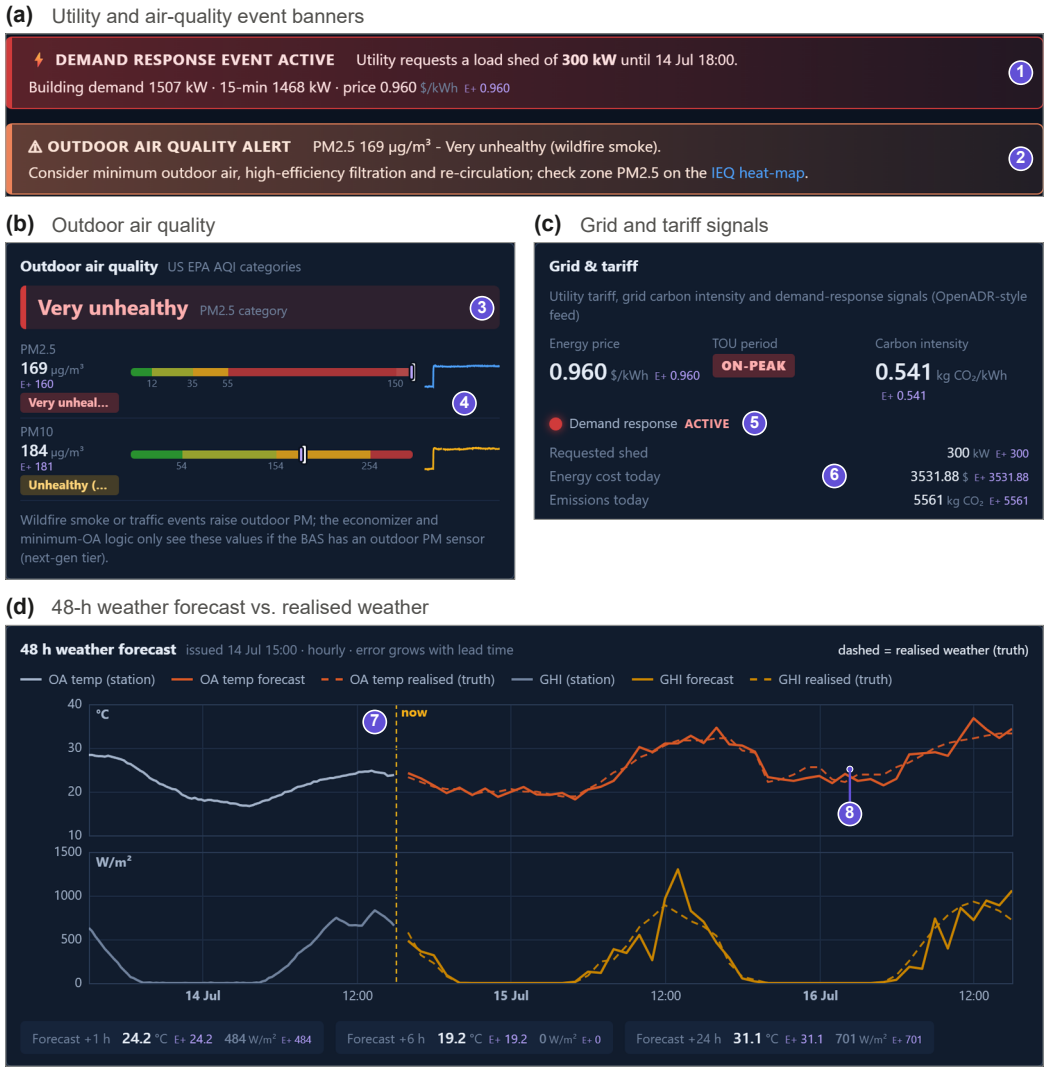

(a) Utility and air-quality event banners
DEMAND RESPONSE EVENT ACTIVE Utility requests a load shed of 300 kW until 14 Jul 18:00.
Building demand 1507 kW · 15-min 1468 kW · price 0.960 $/kWh E+ 0.960
1
OUTDOOR AIR QUALITY ALERT PM2.5 169 µg/m³ - Very unhealthy (wildfire smoke).
Consider minimum outdoor air, high-efficiency filtration and re-circulation; check zone PM2.5 on the IEQ heat-map.
2
(b) Outdoor air quality
Outdoor air quality US EPA AQI categories
Very unhealthy PM2.5 category
3
PM2.5
169 µg/m³
E+ 160
Very unheal...
12 35 55 150
4
PM10
184 µg/m³
E+ 181
Unhealthy (...
54 154 254
Wildfire smoke or traffic events raise outdoor PM; the economizer and minimum-OA logic only see these values if the BAS has an outdoor PM sensor (next-gen tier).
(c) Grid and tariff signals
Grid & tariff
Utility tariff, grid carbon intensity and demand-response signals (OpenADR-style feed)
Energy price
0.960 $/kWh E+ 0.960
TOU period
ON-PEAK
Carbon intensity
0.541 kg CO₂/kWh
E+ 0.541
Demand response ACTIVE
5
Requested shed 300 kW E+ 300
Energy cost today 3531.88 $ E+ 3531.88
Emissions today 5561 kg CO₂ E+ 5561
6
(d) 48-h weather forecast vs. realised weather
48 h weather forecast issued 14 Jul 15:00 · hourly · error grows with lead time
dashed = realised weather (truth)
OA temp (station)
OA temp forecast
OA temp realised (truth)
GHI (station)
GHI forecast
GHI realised (truth)
°C
40
30
20
10
7
now
8
W/m²
1500
1000
500
0
14 Jul
12:00
15 Jul
12:00
16 Jul
12:00
Forecast +1 h 24.2 °C E+ 24.2 484 W/m² E+ 484
Forecast +6 h 19.2 °C E+ 19.2 0 W/m² E+ 0
Forecast +24 h 31.1 °C E+ 31.1 701 W/m² E+ 701